

Decision Making Using Marketing Analytics

MKT 566 · Fall 2026

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USC Marshall School of Business

Introduction

A little about me

- **11th year at USC** (office HOH 332)
- During **2023–25**, worked full-time at **Amazon**, first in Ad Measurement, then in Climate Pledge Friendly
 - Worked closely with data scientists, research scientists, sales teams, and product managers
- Currently part-time at **Airbnb** as *Housing Scholar*
- I run the Real-Estate Analytics Lab (REAL) and its free Substack with Marco Giacoletti

Personal website: <https://dadepro.github.io/>.

A little about me

Research on **online marketplaces and policy**:

- Trust and reputation
- Platform manipulation
- Short-term rentals (Airbnb) and real estate
- Advertising
- Sports betting

My website has links to all my papers. They are also on SSRN.

A little about you

- What is your background?
- What do you expect from this class?
- Have you worked in marketing yet?

The Course

Who to contact

	Instructor	Teaching assistant
Name	Davide Proserpio	Raghav Sarmukaddam
Office	HOH 332	HOH 311
Email	proserpi@marshall.usc.edu	sarmukad@usc.edu
Hours	Tue 2–4 pm	Thu 2:30–4:30 pm

Both by appointment as well.

Organization

Schedule. 11:00 am – 12:20 pm, Tuesday and Thursday, in **JKP 112**

Class website. <https://github.com/dadepro/mkt566>

Syllabus. [mkt566-syllabus-proserpio.pdf](#)

Most lectures involve examples and exercises. Some classes are given over entirely to exercises, discussion, or guest speakers.

I am also planning to have a few guest speakers, very likely from tech companies.

This class is hands-on. Bring your laptop and be prepared to analyze data, write code, and present results.

Readings

Slides, plus open-source reading links provided for each topic.

Open source

- [My course on data storytelling](#)
- [Data Visualization: A Practical Introduction](#)
- [R for Data Science](#)
- [R Markdown](#)
- [R for Marketing Students](#)

Not open source (slides, exercises, and code are)

- [R for Marketing Analytics](#)
- [Python for Marketing Analytics](#)

Analyzing data and coding

- I will use **R and Python**
 - R primarily for data visualization and statistics
 - Python primarily for machine learning
- For assignments you may use either, but following what I do is easier, especially if you are new to coding

IDE suggestions: VS Code (pretty much any programming language, integrates with Codex, Claude, and Copilot), RStudio (R only), Cursor.

Setup

Before the next class, install:

1. **R and/or Python** (Python ships with macOS): [install R](#)
2. **An IDE or editor**
 - RStudio for R
 - [VS Code](#) for essentially any language, including R and Python
3. **An AI coding tool**, whichever you prefer: [Claude Code](#), [Codex](#), [GitHub Copilot](#), or [Cursor](#)

Optional: course materials live on GitHub, so skim [git - the simple guide](#) and the [GitHub quickstart](#), or use the point-and-click [GitHub Desktop app](#).

If this is the **first time** you have heard any of these terms, raise your hand and ask questions.

Use of AI (LLMs)

I **expect** you to use AI in this class. Learning to use it is an emerging skill. Keep the following in mind:

- AI is permitted to help you **brainstorm topics** or **revise work you have already written**
- Minimum-effort prompts produce low-quality results. Refining prompts to get good outcomes takes work
- Proceed with caution: **do not assume the output is accurate or trustworthy**

Hallucinations and mistakes are very common.

For transparency, every student submits a **log of the LLM prompts** used on each assignment.

Why this course matters



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Building Fridayy AI | MERN | AI Agents | Gen AI
1d · 

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AI PMs is the new hot job role. Netflix is paying \$900K for AI PMs. Meta is paying \$1M+.

Here are 8 skills every AI PM should have and a short course to start each one.

1

AI & ML Fundamentals

You don't need to train models, but you must understand how they work, their lifecycle, and their limits (hallucinations, bias, cost).

 Course: AI Product Management Specialization – Duke University (Coursera)

2

Applied AI Patterns & Tooling

Know agentic patterns (ReAct, Plan-and-Execute, CodeAct) and AI stacks (LLM APIs, vector DBs, LangChain, CrewAI).

 Course: IBM AI Product Manager Professional Certificate

3

Data Literacy & Evaluation

Be able to query, inspect logs, run A/B tests, and measure model quality (precision, recall, cost per token).

 Course: Generative AI for Product Managers Specialization

4

AI UX & Product Thinking

Design flows that balance creativity and control. Confidence scores, prompt chaining, guardrails.

 Course: AI for Product Management Course (Pendo)

5

Cross-Functional & Documentation Skills

Translate between engineers, data scientists, and business teams. Write PRDs with clear inputs, outputs, guardrails, and metrics.

 Course: Artificial Intelligence Product Certification (Product School)

6

AI Strategy & Business Acumen

Spot high-ROI AI use cases, evaluate competitors, and model trade-offs (latency, accuracy, cost).

 Course: Mastering Generative AI for Product Innovation (Stanford Online)

7

Experimentation & Deployment Mindset

Own offline + online evaluations. Know when to tweak UX, retrain models, or change data inputs.

 Course: AI Product Management Bootcamp (Maven)

8

Emerging Trends Awareness

Stay ahead on new models (Gemini, Claude, Mistral), multimodal AI, and evolving standards like MCP.

 Course: Elements of AI – University of Helsinki

Source

Why this course matters

- **Budgets follow measurement:** most marketing spend is now digital and measurable, so the marketers who can show what actually works are the ones who win the budget
- **Hiring reflects it:** analytics skills (SQL, R/Python, experiments) appear in a growing share of marketing job postings, from brand management to growth roles (we will look at the job-postings data on Thursday)
- **AI raises the bar, it does not lower it:** LLMs make code and analysis cheap. What stays scarce is judgment: **which question to ask, whether the answer is credible, and what decision follows**

That judgment is what this course trains: framing marketing questions, picking the right method, and turning analysis into decisions.

Grading

Component	Weight
4 individual assignments	60%
Semester-long project (groups of 5–6)	30%
Participation	10%
Total	100%

No final exam. Your grade is the weighted average above.

Assignments

- 4 individual assignments, 60% of the grade
- Roughly **three weeks** for each

Each submission is an **R (Markdown/Quarto) or Python notebook**, converted to HTML or PDF, containing:

- Code, properly commented
- Outputs (figures, tables)
- Your answers to the assignment questions
- The **log of LLM prompts** used

Why a notebook? Reproducibility. Someone else, including future you, has to be able to re-run it and get the same output and results.

The Semester Project

What it is

Analyze a **real-world dataset** to answer one or more marketing questions. For example:

- **Social media sentiment analysis**: how brand perceptions changed over time, which content performs better
- **Customer segmentation**: who are our most valuable segments?
- **Churn prediction and retention**: which users are likely to churn next month?

An example from last year

NETFLIX
CHURN PREDICTION ANALYSIS

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Professor Proserpio

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Exploring Customer Satisfaction Factors for Olist E-Commerce



Driving Pet

Product Success

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Group 8
Final Presentation

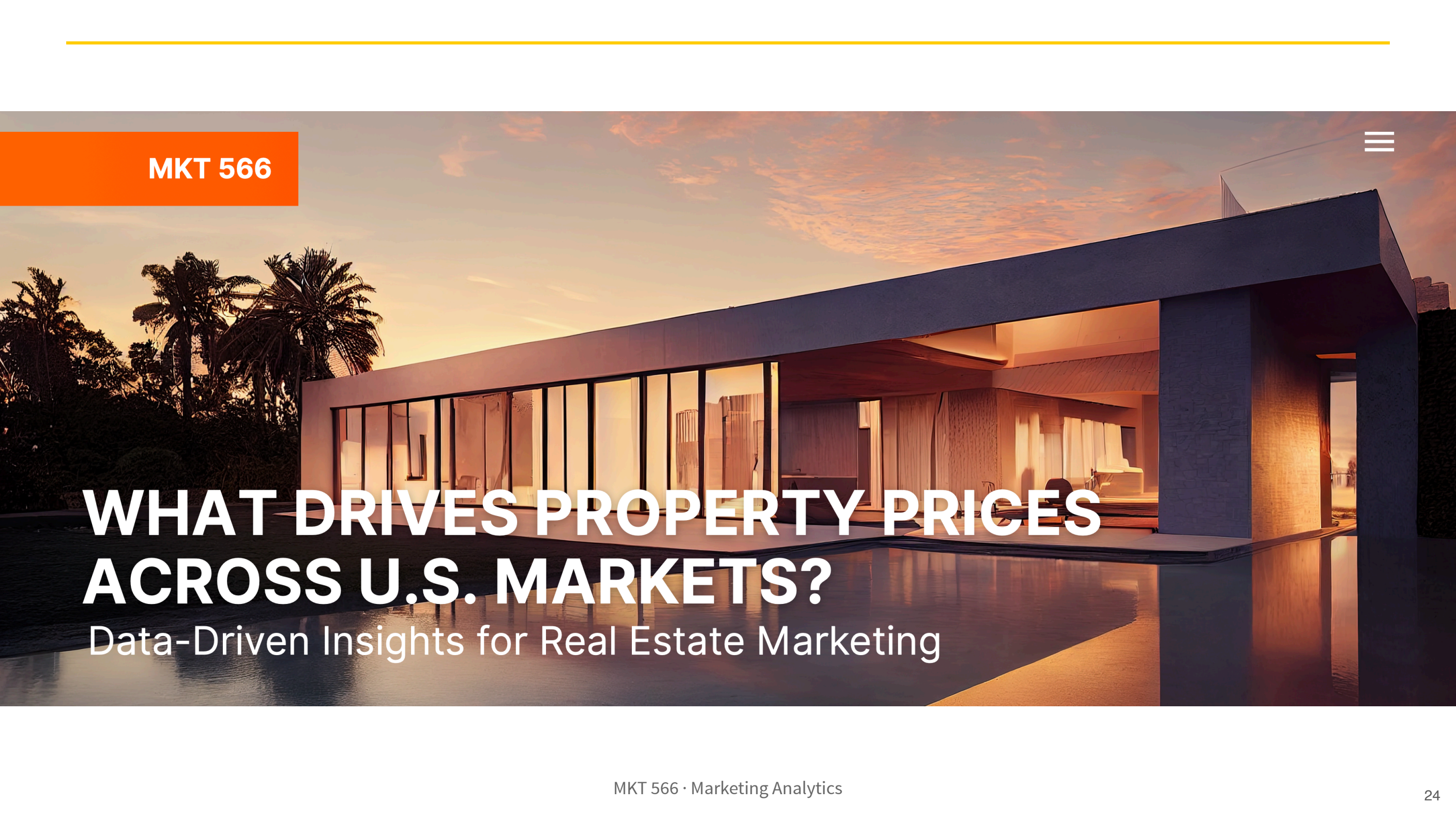
GROUP 5

CUSTOMER SHOPPING MARKETING ANALYSIS

The background is a white canvas with various hand-drawn musical elements. On the left, there is a detailed drawing of an acoustic guitar. Scattered throughout are several musical notes of different sizes and styles, some with stems and flags. In the bottom left, there is a drawing of a concert ticket with the word "Concert" and some scribbled lines. At the top center, there is a small, stylized drawing of a Christmas tree. The entire scene is framed by a thin yellow line at the top.

SPOTIFY PROJECT

What Music To Produce?



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WHAT DRIVES PROPERTY PRICES ACROSS U.S. MARKETS?

Data-Driven Insights for Real Estate Marketing

Examples from a colleague's class

ChatGPT and online content

Using Stack Overflow and YouTube data, students found a decline in Stack Overflow activity for coding topics (Python, Java) alongside an increase in AI-related videos on tech channels.

[Read it](#)

Examples from a colleague's class

Sentiment analysis of Sephora reviews

Using customer review data, students asked whether reviews help identify which Sephora products get featured as *trending* on the website.

[Read it](#)

Examples from a colleague's class

Breaking box office

Using Kaggle movie reviews and Box Office Mojo revenues, students documented a **null result**: customer reviews generally do not explain box office performance, though they may matter for less popular movies.

[Read it](#)

Examples from a colleague's class

Gender effects in a dating app

Using Bumble review data, students found that although more users appear to be male, women rate the app higher than men. A nice application of the `gender-guesser` package.

Where to find data

- Kaggle datasets, e.g. customer churn, shopping trends, Steam
- Yelp open dataset
- Amazon, Google reviews, Steam and more
- Marketing Science papers often ship data and code for replication
- **Collect it yourself** via an API

I have data on **Airbnb, TripAdvisor, Expedia, and Yelp**. Ask me if you want it.

Deliverables and deadlines

Deliverable	Due
Form groups	September 15
Mid-term proposal slides (~15 min incl. Q&A)	October 12 (presented Oct 13, 15)
Notebook with cleaning and analysis (HTML or PDF)	December 3
Final presentation slides (~15 min incl. Q&A)	November 30 (presented Dec 1, 3)
Peer evaluations	December 15

Do you want me to form the groups, or would you rather do it yourselves?

Participation

10% of your grade, and it is not just coming to class.

- This course is an **active learning experience**: participation is graded on the **quality and quantity** of your contributions in each lecture
- I record **attendance** at the beginning of most classes
 - You can miss up to **two sessions** without penalty; more than two will lower your participation grade
 - If you miss a class, getting the notes and handouts is your responsibility
- **Course norms count**: be on time, do not leave early, and no phones or other off-task devices; violations affect your participation grade

Read the syllabus

Everything I have just discussed is in the syllabus. Please **read it at least once**.

Marketing Analytics

What it is

Marketing analytics is the practice of measuring, managing, and analyzing data from marketing activities to optimize business performance.

It covers:

1. Data collection and integration
2. Visualization and reporting
3. Advanced analysis
4. Defining, measuring, and tracking performance (KPIs)
5. Translating insights into decisions

1. Data collection and integration

- **Gathering** information from the web, social media, e-commerce platforms, and elsewhere
- **Unifying** datasets from different sources (online clicks, offline transactions) so they can answer a marketing question together

2. Visualization and reporting

Creating figures, dashboards, and reports that present insights to stakeholders:

- Advertising effect on sales
- Eco-labeling program growth
- Relationships between variables

3. Advanced analysis

Attribution modeling (causality)

Which channels or campaigns deserve credit for conversions?

Recommendation and segmentation (clustering)

Group customers by behavior or demographics to tailor messaging; identify products consumers are likely to buy; product placement

Predictive modeling (regression, machine learning)

Forecast campaign performance; predict churn; identify fraudulent activity such as fake reviews, accounts, and clicks; forecast prices

4. KPIs

Metric family	Examples
Acquisition	Cost per click, cost per acquisition
Engagement	Click-through rate, time on site
Conversion	Lead-to-customer rate, average order value
Retention	Churn rate, customer lifetime value
Incrementality	How much of the sales lift is <i>attributable</i> to the campaign

5. Translating insights into decisions

Turning analysis into action:

- **Reallocating** budget to top-performing channels
- **Optimizing** messaging and creative
- **Changing** the landing page
- **Identifying** and leveraging new trends
- **Optimizing** prices

In a nutshell

By systematically applying statistical methods and data-driven storytelling, marketing analytics enables organizations to **understand what works**, **identify growth opportunities**, and **maximize their return on marketing investment**.

The Course Content

What we will cover

- Data analysis and visualization
- Regressions
- Clustering and recommendations
- Classifiers
- Causality: experiments and observational data
- **LLMs and agents**: new challenges for brands (e.g. search), and how firms are leveraging them

What we will *not* cover

- **This is not a programming course.** Learning to code is mostly up to you and your LLM
- **This is not an advanced statistics, econometrics, or ML methods course.** The emphasis is on **data-driven decision-making**, so we stay at a high level conceptually

I assume you are familiar with **basic statistics and probability**. I can post review slides if that would help.

Questions?